

To understand if the technology is sufficiently mature, we must look at all the components that make a home automation system.

There are three main components in home automation technology:

1. Your device
 - a. Device is the electronic hardware that you wish to control using your home automation system.
2. Automation Software
 - a. Automation software is the logic that decides what action must be performed by a particular device and helps us to control the device.
 - b. Parts of the software is developed using programming languages like C, and part of it can be developed in scripting languages like Python for specialised hardware.
3. Automation hardware
 - a. Automation hardware is an electronic device between the software and the device that you wish to control.
 - b. Hardware sends the appropriate signal to the device depending on input it gets from sensor or you as a user. For example, some signals an automation hardware can send to an air-conditioner are to increase or decrease the temperature of a room.
 - c. The signal to be sent to your device is decided by the automation software that runs on the automation hardware.

There are a large number of developers who are contributing towards improving existing software and making it suitable for automation using various open-source communication protocols. However, the lack of a unified security framework has plagued the IoT segment for quite some time. And this is being exploited by hackers. A recent DDoS (Distributed Denial of Service) attack in the third week of October 2016 on popular services like Twitter, Spotify and GitHub among others was suspected to be carried out by exploiting the vulnerabilities in IoT (Internet of Things) software which is very much the same as home automation software.

We can safely say that home automation is currently in its nascent stages having a great room from improvement.



Nest home temperature control hardware